

hybrid-text-retrieval-toolkit: BM25 + dense + ColBERT-style hybrid retrieval on BEIR

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1 Abstract

We present `hybrid-text-retrieval-toolkit`, a production-friendly hybrid retrieval harness that evaluates BM25, dense bi-encoder, ColBERT-style late-interaction, and cross-encoder reranking on the BEIR benchmark suite (Thakur et al., 2021). We report results on the SciFact sub-

dataset (5,183 documents, 809 queries with qrels), showing that the BGE-small dense retriever wins on this corpus ($nDCG@10 = 0.757$) over BM25 ($nDCG@10 = 0.659$) and over the equal-weight RRF fusion ($nDCG@10 = 0.734$), confirming the expected RRF failure mode when one base is much stronger than the other. The toolkit ships six distinct visualization types and is structured to make adding a new retriever or a new BEIR dataset a single-file change.

2 1. Background

Hybrid retrieval (the combination of lexical and dense methods) is the dominant pattern in production information retrieval today. It works because the two component methods make complementary errors: BM25 is exact-match-precise but cannot do soft semantic matching, dense bi-encoders do soft matching but lose information when the document is pooled to a single vector. ColBERT (Khattab & Zaharia, 2020) sits between them: per-token embeddings preserve where the match happened but at higher storage cost.

This toolkit is the cheapest way to compare those three approaches plus reranking on a new corpus. It uses BEIR-format datasets so benchmarking against the published numbers is direct, and it ships six chart functions that answer different questions about the same per-query data so the leaderboard isn't the whole story.

3 2. Related Work

BEIR. Thakur et al. (2021) introduced BEIR as the standard zero-shot IR benchmark; we use the same HF mirror under BeIR/<name> with one helper that converts to our internal types.

BM25. Robertson & Walker (1994). rank-bm25 with $k1=1.5$, $b=0.75$.

Dense bi-encoders. BGE family (Xiao et al., 2024); BGE-small-en-v1.5 is our default for laptop tractability.

Late interaction. ColBERT (Khattab & Zaharia, 2020) and ColBERTv2 (Santhanam et al., 2022). Our implementation is the brute-force MaxSim variant (no PLAID indexing); for corpora over 100K documents the PLAID approach is necessary.

RRF. Cormack et al. (2009). Default $k=60$, over-fetch=4.

4 3. Method

4.1 3.1 Architecture

```
src/hr/
  types.py          Doc, Query, Hit, Qrels
  indexes/
```

base.py	Index ABC
bm25.py	rank-bm25 + title-aware tokenization
dense.py	BGE-small + FAISS IndexFlatIP
late_interaction.py	per-token MaxSim (brute force)
encoders/cross.py	cross-encoder rerank
fusion/	
rrf.py	Reciprocal Rank Fusion
linear.py	score-level fusion with weights
scoring/	
metrics.py	nDCG, Recall, MRR, MAP
runner.py	orchestrate + write artifacts
plots.py	six distinct chart types
data.py	BeIR HF loader
cli/main.py	typer: bench, plots, report

4.2 3.2 ColBERT-style MaxSim

We embed each document into $(n_patches, d)$ per-token embeddings using a sentence-transformers backbone (all-MiniLM-L6-v2 by default). The score for a (query, document) pair is:

$$\text{score}(q, d) = \sum_{i \text{ in query tokens}} \max_{j \text{ in doc tokens}} \langle q_i, d_j \rangle$$

This is the original ColBERT MaxSim formulation. We do not implement the centroid + IVF pruning from ColBERTv2 because brute-force is fine at SciFact scale (~5K docs).

4.3 3.3 Six chart types

1. **nDCG@k curves** — how each index's quality scales with k
2. **Recall-precision tradeoff** — the classic IR curve
3. **Per-query nDCG distribution** — wide tail vs concentrated wins
4. **Quality vs speed scatter** — Pareto frontier
5. **Build cost vs quality** — index amortization
6. **Per-(index, dataset) heatmap** — for multi-dataset runs

5 4. Data

The headline run uses BEIR/SciFact: 5,183 scientific abstracts as the corpus and 809 queries with relevance judgments. Other BEIR sub-datasets (NF-Corpus, FiQA, TREC-COVID, etc.) are one-flag swaps; we did not run the full BEIR sweep in this iteration.

6 5. Evaluation Setup

Standard BEIR metrics: nDCG@k, Recall@k, MRR@k, MAP@k at $k \in \{1, 3, 5, 10, 20\}$. Build time and per-query latency are reported alongside.

Hardware: Apple M-series, CPU.

7 6. Results

Headline run on BEIR/SciFact (809 queries, 5,183 docs):

index	nDCG@10	Recall@10	MRR@10	MAP@10	build (s)	QPS
bm25	0.659	0.781	0.626	0.615	0.28	148
dense (BGE-S)	0.757	0.871	0.724	0.715	66.07	102
rrf(bm25+dense)	0.734	0.851	0.703	0.692	49.23	56

Three findings:

1. **Pure dense wins on SciFact.** This is the opposite of the cuad finding in our sister project (legal-retrieval-benchmark). SciFact’s queries are short scientific claims against ~5K abstracts; the dense encoder’s representation advantage compounds, and BM25’s vocabulary mismatch dominates.
2. **RRF sits between the bases, not above.** With one base much stronger than the other, equal-weight rank fusion drags the stronger one down. This is the expected RRF failure mode.
3. **Build cost is BM25’s structural advantage** (0.28s vs 66s for dense). For high-churn corpora this matters a lot.

8 7. Ablations

Pending; planned items: BM25 (k1, b) sweep, dense (truncation, pooling) sweep, RRF (k, over-fetch) sweep, ColBERT-style MaxSim on a small subset.

9 8. Discussion

The juxtaposition with legal-retrieval-benchmark is the most useful take. Same harness, two corpora, opposite winners: dense wins SciFact, BM25 wins CUAD. The mechanism is the same in both directions: dense encoder representation quality vs document length / truncation. On short documents dense wins; on long structured documents BM25 wins. The corpus characteristics drive the answer, not the abstract method.

10 9. Limitations

1. **Brute-force late interaction.** ColBERT-style runs $O(n_docs \times patches \times patches)$ per query; fine at 5K docs, breaks at 100K.
2. **Single dataset.** Only SciFact is in the published results; full BEIR sweep is queued.
3. **No domain encoders.** BGE-small is general English; a domain-tuned encoder typically lifts dense by 5-15 points.

11 10. Future Work

- ☐ Full BEIR sweep (15+ datasets)
- ☐ PLAID-style centroid indexing for ColBERT
- ☐ Per-dataset BM25 (k1, b) tuning
- ☐ Adaptive routing (BM25 first, escalate to dense if margin low)
- ☐ Domain encoders (Legal-BERT, BioBERT, etc.) per BEIR subset

12 11. References

- Cormack, G. V., Clarke, C. L. A., & Buettcher, S. (2009). *Reciprocal Rank Fusion outperforms Condorcet and individual rank learning methods*. SIGIR.
- Khattab, O., & Zaharia, M. (2020). *ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT*. SIGIR.
- Robertson, S. E., & Walker, S. (1994). *Some simple effective approximations to the 2-Poisson model for probabilistic weighted retrieval*. SIGIR.
- Santhanam, K., et al. (2022). *ColBERTv2: Effective and Efficient Retrieval via Lightweight Late Interaction*. NAACL.
- Thakur, N., et al. (2021). *BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models*. NeurIPS.
- Xiao, S., et al. (2024). *C-Pack: Packed Resources For General Chinese Embeddings*. SIGIR.

13 Appendix A. Reproducibility

- Repo: Akshitha024/hybrid-text-retrieval-toolkit, MIT.
- SciFact run: `make bench DATASET=scifact && make plots`.
- All per-query results in `results/scifact__<index>__runs.jsonl`.
- All six charts in `results/figures/`.
- Test artifacts in `docs/test_results/`.